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An overview of Unmanned Surface Vehicles: Methods, practices, and applications ☆

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Abstract

Over the past few decades, Unmanned Surface Vehicles (USVs) have transitioned from concept to reality at an impressive pace. This rapid development is fueled by advances in technology and emerging demands in science, engineering, and military applications. The emphasis on USVs has evolved from purely research-focused activities to practical commercial and engineering applications. This paper presents insights gained from a decade of USV development, providing an overview of the latest methodologies, design practices, and applications. We review and compare various USVs currently in use, including those for research and field applications. In addition, we categorize and update the state-of-the-art navigation, guidance, and control methodologies that form the core capabilities of USVs, with a particular emphasis on the growing trend of learning-based and data-driven systems. As a crucial aspect, we also discuss the architectural design, implementation, and field testing of USVs. Finally, we explore their applications in three key areas: military operations, oceanographic observation, and intelligent waterborne transportation. Given that technological and practical challenges for USVs remain, we outline considerations and highlight relevant future directions.

Introduction

Rooted in the field of autonomous mobile robotics and more broadly in unmanned systems, Unmanned Surface Vehicles (USVs) have rapidly developed globally in recent years, leveraging multidisciplinary robotics technologies. USVs are a category of marine vessels designed to operate on the water's surface without a crew onboard. They can be remotely operated or capable of autonomous navigation, and they vary widely in size, shape, and functionality, from small boats to larger ships (Liu et al., 2016, Manley, 2008). Note that the term of "Uncrewed Surface Vehicles" (Handegard et al., 2024) is gaining popularity promoting more inclusive language in contexts aiming for gender neutrality. Both terms "unmanned" and "uncrewed" are acceptable and interchangeable based on preference, context, or audience. We will still use the word "unmanned" in the paper.

In the literature, several related terms with USVs are frequently encountered, such as Autonomous Surface Vehicles (ASVs) (Wei, Sun, Chen, & Shi, 2021), Unmanned Marine Vehicles (UMVs) (Zhang, Ye, Chen, & Wang, 2020), waterborne Autonomous Guided Vessels (waterborne AGVs) (Zheng, 2016, Zheng et al., 2022), and Maritime Autonomous Surface Ships (MASS) (Bakdi, Glad, & Vanem, 2021). However, these terms do have distinct meanings. ASVs specifically emphasize the ability to navigate and perform tasks completely independently, without human intervention. UMVs encompass both unmanned surface and underwater vehicles, while waterborne AGVs focus on applications in transportation. MASS typically refers to larger vessels that may have a crew onboard (IMO, 2018). This review will cover the latest developments relevant to USVs.

The development of USVs can be traced back to World War II, during which they were equipped with firearms to carry out remote attacks against enemy targets (Corfield & Young, 2006). In 1946, the United States utilized USVs to collect water samples from nuclear explosion zones (navy, 2007). In the 1960s, USVs were further employed to perform remote mine-sweeping operations, significantly reducing the risk to human personnel (navy, 2007). Since the late 1990s, USVs have experienced rapid development in both military and civilian domains. In military applications, USVs are primarily used for mine countermeasures, anti-submarine warfare, special operations, and electronic warfare. In the civilian field, USVs have increasingly assumed significant roles in environmental monitoring, maritime patrol operations, waterborne engineering, scientific research. They offer advantages over traditional crewed vessels in terms of safety, comfort, flexibility, and efficiency (Wang & Fu, 2022). USVs are also viewed as pivotal innovations poised to transform the maritime sector (Negenborn et al., 2023). Currently, many countries and organizations are vying for leadership in this area, with European, Asian nations, and the US showing notable activity. Norwegian company Maritime Robotics has received approval to establish the world's first freight route for uncrewed cargo vessels (Maritime Robotics, 2023). The Chinese company Yunzhou has successfully demonstrated a swarm of 56 USVs at sea (China Industrial Press, 2023). Dutch researchers have commented in Nature that the industry should now concentrate on USVs (Negenborn et al., 2023). These events have taken place recently. Several strategic-level projects, non-profit alliances, and policy roadmaps focusing on USVs have been developed over the past decade, as detailed in Table 1. European and Asian countries have predominantly concentrated on civilian applications of USVs, such as autonomous shipping. Meanwhile, the US has led in technological development through various military projects but lags behind European countries in autonomous shipping (Pribyl, 2021). Prominent USV competitions include the Maritime RobotX Challenge, World Robotics Sailing Championship, Microtransat Challenge, Shell Ocean Discovery XPRIZE, Robonation Competition, RoboBoat Competition, Navy's Advanced Naval Technology Exercise (ANTX), Njord International Student Challenge, Zhuhai Wanshan International Intelligent Vessel Competition, National Ocean Vehicles Competition, and the US Defense Advanced Research Projects Agency (DARPA) competitions.

Significant efforts from both academia and industry have been directed toward the methods, practices, and applications of USVs. In recent years, there has been substantial progress in USV research. Numerous review articles (Barrera et al., 2021, Campbell et al., 2012, Goerlandt, 2020, Li, Zhang et al., 2024, Liu et al., 2023, Liu et al., 2016, Na et al., 2015, Peng, Wang et al., 2020, Qiao et al., 2023) have been published to update developments and outline potential challenges regularly. Some reviews focus on specific aspects of USVs, such as collision avoidance (Campbell et al., 2012, Yin and Xiang, 2025), overactuated USVs (Nađ et al., 2015), safety considerations (Goerlandt, 2020), coordinated control (Peng, Wang et al., 2020), and deep learning applications (Qiao et al., 2023). Other reviews explore future directions for transitioning USVs from research concepts to commercial products (Barrera et al., 2021) and their use in inland shipping (Liu et al., 2023). Notably, Liu et al. (2016) provides a comprehensive review of approximately 200 papers on USVs up to 2016. Over the past eight years, this paper alone has been cited nearly 900 times, indicating significant progress in the USV field since then and underscoring the topic's popularity. The extensive research has been categorized by platforms, applications, control techniques, and more, reflecting the dynamic growth of interest and development in USVs.

However, existing review works primarily focus on the algorithmic and methodological aspects of USVs. While guidance, navigation, and control (GNC) algorithms are critical, recent years have also witnessed successful applications of various commercial USV products (China Industrial Press, 2023, Li, Zhang et al., 2024, Maritime Robotics, 2023). Building on methodological advancements from the past several decades, the focus of USV research and development is increasingly shifting toward practical validation and application-oriented capabilities. Practical validation of USVs carefully considers the interaction of power and data flows between various subsystems, integrating mechanical design, software development, circuit modules, payload selection, and USV-to-shore communications. The complex marine environment poses unique challenges to the reliability of USV systems and to validation testing procedures. Conversely, the ability to operate autonomously in uncertain and unstructured marine environments without human intervention holds significant value across various applications. The development of application-oriented capabilities in specific engineering, scientific, and military contexts still requires further enhancement. Undoubtedly, the future progress of USVs will depend on the combined advancement of methods, practices, and application capabilities. There is a pressing need for a review that addresses the state-of-the-art in USV research across these dimensions.

This paper offers a review of the latest advances in USVs, focusing on methodologies, design practices, and applications. Drawing on the insights gained from over a decade of experience in the field, we assert that the three aspects — methods for achieving autonomy, practices for validation, and application capabilities — are equally crucial for the future progress of USVs. We begin by reviewing existing typical USVs, followed by a survey that emphasizes the following three aspects:

- The state-of-the-art GNC methods, which include both classical techniques that form the foundation of USV autonomy and emerging learning-based technologies;
- Systematic design, implementation, and testing practices for USVs;
- Application capabilities of USVs across three distinct fields: military operations, intelligent waterborne transportation, and oceanographic observation.

At the end of the paper, we highlight the technological and practical challenges that remain for USVs and provide perspectives on future research directions. To the best of the authors' knowledge, this is the first review that summarizes the relevant methods, practices, and applications of Unmanned Surface Vehicles (USVs). This review aims to provide the USV research and industry communities with an overview of the field, enhance research efficiency, and accelerate the development of practical USV application capabilities. Leveraging our academic expertise in USVs, we compiled a list of literature references through a thorough search of official websites and reputable academic databases such as Google Scholar and Web of Science. During the search, relevant keywords, e.g., "Unmanned Surface Vehicles (USVs)", "navigation, guidance and control of USVs", "USV tests", "USV applications", and "overview of USVs", related to the review are utilized. We prioritize citing peer-reviewed articles, conference papers, authoritative reports, and recent official website introductions. These literature references offer readers valuable sources on USV research and help establish a credible foundation for our perspectives. The specific literature selection criteria include: Relevance to the core themes of our research; Impact factor of the journals and credibility of the authors; Contributions to existing knowledge that support or contrast with our viewpoints.

The remainder of this paper is organized as follows. Section 2 provides an update on typical USV platforms and essential USV components. Sections 3 through 5 review the state-of-the-art GNC methods for USVs. Section 6 discusses the work and experiences related to the system implementation and validation of USVs. The applications of USVs in three representative fields are detailed in Section 7. Finally, Section 9 concludes the paper and presents future perspectives.

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Section snippets

State-of-the-art USV platforms

A decade ago, much of the work on USVs focused primarily on pure research purposes (Zheng, Negenborn, & Lodewijks, 2013). Small-scale prototypes were constructed and tested within laboratory settings (Skjetne, 2005, Zheng, 2016). However, with the maturation of methodologies and technologies, we are now witnessing the development of USVs with higher levels of automation that are being practically applied across various fields. This section provides an update on the existing USV platforms. …

USV navigation methods

Navigation refers to the decision-making process that ensures the safe operation of vessels, encompassing adherence to regulations, managing emergency situations, and maneuvering through complex environments (Fossen, Second edition, 2021). The navigation system enables the USV to gather information such as position, speed, water depth, environmental data, and other marine-specific measurements. In manned ships, the navigation task is primarily performed by the Officer of the Watch; in contrast, …

USV guidance methods

Guidance refers to the process of steering the USV along a designated path to reach specific waypoints (Shneydor, 1998). This task typically involves two steps: (1) planning a path on the navigation map for the USV, and (2) maneuvering the USV along the planned path to guide it toward the designated waypoints. …

USV motion control methods

For USVs, the control module functions as a closed-loop feedback subsystem that employs sophisticated algorithms to achieve desired behaviors. The control system utilizes filtered outputs from the navigation module, compares these outputs to reference signals from the guidance module, computes corrective inputs — often based on the USV's dynamics model — and applies these inputs to the USV system. Generally, USV control tasks can be categorized into three types: setpoint regulation, path …

Practices

In addition to key research methodologies related to GNC, systematic design practices are crucial for transitioning USVs from theoretical research concepts to field operations and ultimately to mature commercial products. This section summarizes the insights gained by the authors from practical designs and testing of USVs. Specifically, we will elaborate on the core hardware components of USVs, the generic USV software architecture, and the testing procedures for USVs. …

Applications

The applications of USVs, primarily small in size, initially emerged in the military sector and later expanded to areas such as ocean science and mineral detection. Subsequently, larger USVs have been developed for autonomous waterborne transportation (Grose, 2022, Zheng, 2016). …

Future perspectives

The past decade has seen rapid development and growing applications of USVs. The widespread adoption of fully autonomous USVs presents a range of specific and entirely new challenges, necessitating further technological and practical advancements. In this paper, we outline several key technical issues associated with USVs and provide recommendations for future work in these areas. …

Conclusions

This paper provides an overview of the ongoing research on USVs over the past decade. As USVs have progressed from concept to reality, they have become increasingly autonomous, versatile, and reliable for a variety of scientific, engineering, and military applications. The core components that underpin USV autonomy — namely navigation, guidance, and control methods — have reached a relatively mature stage. However, emerging technologies, such as deep learning algorithms, continue to yield …

CRediT authorship contribution statement

Huarong Zheng: Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Chenguang Liu:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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